

Shapiro-Wilk Test

Raúl Fauste

September 2026

1 Introduction

The Shapiro-Wilk test is a parametric test for statistical hypothesis testing used to test if a poblation is normally distributed. When using the test for algorithm analysis, the poblation under study is the difference between the objective function values of both algorithms. In these cases, the goal is to test if the difference is normally distribute, in order to know how to proceed for a deeper analysis.

2 Test Procedure

The data is represented as $x_1, x_2, \dots, x_n$ and the null hypothesis assumes that is normally distributed. The test statistic is

$$W = \frac{(\sum_{i=1}^n a_i x_{(i)})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

where

- $x_{(i)}$ with parentheses enclosing the index i , is the i th-smallest number in the sample (once the sample x is ordered).
- $\bar{x} = \frac{x_1 + \dots x_n}{n}$

The coefficients a_i are given by:

$$(a_1, \dots, a_n) = \frac{m^t V^{-1}}{C}$$

and the vector $m = (m_1, \dots, m_n)^t$ is made of the expected values of the order statistics of independent and identically distributed random variables sampled from the standar normal distribution. Also, V is the covariance matrix of those normal order statistics. Usually, this coefficients values are obtained from fixed tables.

As a statistical test, it relies on a null hypothesis (H_0) and an alternative hypothesis (H_1):

- H_0 : Assumes the data is normally distributed.
- H_1 : Assumes the data is not normally distributed.

The goal of this test is to conclude that the data is not normally distributed. This test is normally used in order to proceed with a more complex test for a deeper analysis. In algorithm analysis, it is the first test applied, then if the data is normally distributed may be a t-test is executed or otherwise, Wilcoxon Signed-Rank test, for example.

3 Interpretation

After performing this test, the statistic $W \in [0, 1]$. The closer the value is to 1, means that the population is normally distributed. Otherwise, the population is not normally distributed. Running the test on any software will also report the value of the associated p -value. If we establish a confidence value α depending on the p -value obtained we are able to reject (or not) the null hypothesis.

The relation between the statistic W and the corresponding p -value, is often calculated using tables (where data is obtained from simulations).